

Source Device & Pin	Signal Name	Destination Device & Pin	Signal Function		5000 = 5vDC Ground
ESP32 GPIO 23	MOSI	SN74AHCT125 Pin 9 (3A)	3.3V Serial Data Input		5500 = 5vDC
ESP32 GPIO 18	SCK	SN74AHCT125 Pin 2 (1A)	3.3V Shift Clock Input		6110 = 110vAC Hot
ESP32 GPIO 5	CS/SS	SN74AHCT125 Pin 5 (2A)	3.3V Latch Clock Input		6100 = 110vAC Neutral
ESP32 GPIO 4	PWM/EN	SN74AHCT125 Pin 12 (4A)	3.3V Output Enable Input		7500 = Serial In/Out
SN74AHCT125 Pin 8 (3Y)	SER_5V	TPIC6B595N Pin 3 (SER IN)	5V Converted Data		8100 = 5v Enable
SN74AHCT125 Pin 3 (1Y)	SRCLK_5V	TPIC6B595N Pin 8 (SRCK)	5V Converted Shift Clock		8200 = 5v Latch
SN74AHCT125 Pin 6 (2Y)	RCLK_5V	TPIC6B595N Pin 9 (RCK)	5V Converted Latch Clock		8300 = 5v Clock
SN74AHCT125 Pin 11 (4Y)	OE_5V	TPIC6B595N Pin 12 (G)	5V Converted Output Enable		
TPIC6B595N Pin 18	SER_OUT	TPIC6B595N #2 Pin 3 (SER IN)	Board-to-Board Serial Cascade		
TPIC6B595N Pins 4–7, 14–17	DRAIN0–7	SSR Pin 2 (Input -)	Low-Side Sink Drive to Relays		
+5V Rail	+5V	SSR Pin 1 (Input +)	Common Anode Supply to Relays		
+5V Rail	+5V	TPIC6B595N Pin 13 (CLR)	Pulls Clear High (Disables Clear)		
System Ground (0V)	GND	TPIC6B595N Pins 1, 2, 10, 11, 19	Combined PGND1–4 & LGND		
System Ground (0V)	GND	SN74AHCT125 Pins 1, 4, 7, 10, 13	Ground Reference & Active OE		